

SatQuery-Edge

Offline Satellite Intelligence Investigation Report
SIH 26167 · ISRO · Prototype · Offline Mode

Investigation ID	SQE-20260902-JHSSHB
Timestamp	2026-09-02 18:13:27
Intent	single_image_vqa
Tools Used	vqa, grounding, geospatial

Query

describe the image

Final Answer

Scene analysis complete.

=== Llama-3.2-1B-Instruct Edge SLM Analysis ===

Scene contains extensive vegetation or agricultural fields, dense built-up structures, road networks or linear structures.

Illumination appears normal.

Approximately 796 distinct structural regions detected.

Evidence Score: 82/100.

Evidence Score

Component	Code	Value	Weight
Model Agreement	M	0.880	0.30
Sensor Reliability	S	0.840	0.20
Spatial Consistency	P	0.887	0.20
Temporal Consistency	T	0.820	0.15
Input Quality	I	0.598	0.15
FINAL SCORE		82.2 / 100	

Validation

Status: **PASS** ✓

- 1 image(s) detected — count OK.
- Image 1: Quality OK (mean=120, std=42.1).

Detected Regions & Approximate Coordinates

#	Label	Confidence	Area (km²)	Demo Georeferencing
1	candidate regions of interest	0.86	138.471	Yes

Specialist Tool Outputs

VQA — SUCCESS — 3 region(s)

```
=== Llama-3.2-1B-Instruct Edge SLM Analysis ===
Scene contains extensive vegetation or agricultural fields, dense built-up structures,
road networks or linear structures.
Illumination appears normal.
Approximately 796 distinct structural regions detected.
```

GROUNDING — SUCCESS — 1 region(s)

```
Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'regions of
interest'.
Highest confidence: 0.86.
```

GEOSPATIAL — SUCCESS — 0 region(s)

```
Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 138.47 km².

■ Approximate demo georeferencing applied.
Demo center: (20.296°N, 85.824°E)
Coordinates are NOT derived from real GeoTIFF metadata.
```

Execution Trace

```
[18:13:27] QUERY RECEIVED: describe the image
[18:13:27] ROUTER DECISION: Intent: single_image_vqa
Selected tools: vqa, grounding, geospatial
Images required: 1
[18:13:27] VALIDATION: Status: PASS
1 image(s) detected – count OK.
Image 1: Quality OK (mean=120, std=42.1).
[18:13:27] TOOL EXECUTING: vqa → started
[18:13:27] TOOL OUTPUT (vqa): Success – 3 region(s) detected
=== Llama-3.2-1B-Instruct Edge SLM Analysis ===
Scene contains extensive vegetation or agricultural fields, dense built-up structures,
road networks or linear structures.
Illumination appears normal.
[18:13:27] TOOL EXECUTING: grounding → started
[18:13:27] TOOL OUTPUT (grounding): Success – 1 region(s) detected
Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'regions of
interest'.
Highest confidence: 0.86.
[18:13:27] TOOL EXECUTING: geospatial → started
[18:13:27] TOOL OUTPUT (geospatial): Success – 0 region(s) detected
Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 138.47 km².
[18:13:27] EVIDENCE FUSION: Score: 82.2/100
M=0.880
S=0.840
P=0.887
[18:13:27] FINAL SCORE: 82/100
[18:13:27] ANSWER: Scene analysis complete.

=== Llama-3.2-1B-Instruct Edge SLM Analysis ===
Scene contains extensive vegetation or agricul
```

Tool Versions

Component	Version
SatQuery-Edge Prototype	1.0.0-SIH26167
Python OpenCV	4.11.0
NumPy	1.26.4
Streamlit	1.62.0
Report Engine	ReportLab PDF
Mode	OFFLINE — No external APIs

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. — SIH 26167 · ISRO · SatQuery-Edge